GUVI – Data Analyst Trainee Task

Sales Data Analysis & Insights Report

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1. INTRODUCTION:

The objective of this project is to analyze the given dataset to derive actionable insights that can support data-driven decision-making. This analysis has been carried out as part of the Data Analyst Trainee assignment for GUVI, aimed at evaluating the ability to handle, visualize, and interpret real-world data effectively using analytical tools such as Microsoft Power BI.

The dataset provided contains detailed information related to coupon transactions, user behavior, and mode of purchase, which are essential to understanding customer engagement and promotional effectiveness. The goal is to identify meaningful trends, segment user types, and evaluate the performance of various coupon types across different purchase modes.

Through this report, the focus is to:

- Perform data cleaning and transformation for accurate reporting.
- Develop interactive dashboards using Power BI to visualize key metrics.
- Calculate custom measures (such as User Value, Repeat Users, and One-time Users).
- Derive insights that highlight user purchasing behavior and coupon performance.

The findings from this analysis can help optimize marketing strategies, improve customer retention, and guide decision-makers in designing more effective promotional campaigns.

2. DATASET OVERVIEW

The dataset consists of 9514 records and 11 attributes, covering daily sales data for two months (June and July 2024). It provides detailed insights into transactions, user behavior, sales sources, and payment preferences, making it suitable for trend and performance analysis in Power BI.

KEY ATTRIBUTES:

- User ID, Product Code, Sales Date – identify unique users, products, and daily transactions.
- Payment Status, Payment Mode, Transaction Bank – represent payment completion, mode of payment, and financial source details.
- Coupon Code, Source – track promotional code usage and sales channel effectiveness.
- Product Amount with GST, Currency Code – indicate total transaction value and the currency type.
- Lead Registered Time – captures the timestamp of lead creation, useful for conversion and time-based analysis.

The dataset was cleaned and transformed in Power BI Power Query, ensuring data quality by removing duplicates, correcting date formats, and replacing null values (e.g., “Unknown” for missing entries) to enable reliable visualization and insights.

3. DATA CLEANING & PREPARATION

The dataset was processed and refined using Power Query in Power BI to ensure completeness and consistency before analysis. Duplicate and unnecessary rows were removed to maintain data integrity. Missing numeric values in the Product_Amount_with_GST column were replaced with the average value of 1543.02, ensuring no sales record was left incomplete.

Text-based missing values in fields such as Coupon Code, Payment Mode, and Transaction Bank were filled with the placeholder “Unknown” to preserve categorical consistency. Column names were standardized for clarity by removing spaces and maintaining a uniform naming format. Both Sales Date and Lead Registered Time were converted into appropriate Date/Time formats for accurate time-based visualization.

After cleaning, the dataset contained **8909 rows and 11 attributes**, resulting in a well-structured dataset ready for visualization and actionable insights in Power BI.

4. DASHBOARD DESIGN & VISUALIZATION

An interactive sales analysis dashboard was developed in **Power BI** to uncover actionable insights from the two month sales given sample dataset.

Total Sales KPI:
This visualization represented the total revenue generated using the sum of Product Amount with GST over a two month period. It offered a clear snapshot of overall business performance and served as a key indicator for assessing sales growth and revenue trends.

Customer Reach KPI:
This metric was derived using a DAX measure that calculated the distinct count of User ID to determine the total number of unique customers. It provided valuable insights into customer reach and engagement levels, reflecting the brand’s ability to attract and retain buyers.

Average Sales per User (Bar Chart):
This measure was calculated by dividing total sales by the number of unique users to find the average spending per customer. The visualization compared spending patterns across different sales sources, revealing which channels contributed the most valuable and high-spending customers.

Top Performing Products (Clustered Column Chart):
This visualization displayed total sales for each product, sorted in descending order to highlight the **top 10 revenue generating products**. It clearly identified which products contributed the most to overall sales, providing direction for inventory focus and promotional strategies.

Sales Performance by Source (Pie Chart):
This visualization illustrated the share of total sales generated from different sources such as Direct, Referral, and Affiliate. The chart clearly showed that Direct sales contributed the highest proportion of overall revenue, while Referral and Affiliate channels had smaller shares. This insight highlighted opportunities to strengthen underperforming channels through improved marketing strategies or partnership programs to achieve a more balanced sales distribution.

Repeat Users vs One-Time Users (Donut Chart):
This visualization compared the proportion of repeat customers to one-time buyers, providing clear insights into customer loyalty and retention trends. It highlighted the need to strengthen engagement strategies that encourage repeat purchases and long-term customer relationships.

Sales Influence by Coupon Code and Payment Mode (Matrix):
This visualization examined the relationship between payment methods and coupon usage, offering insights into how discounts and preferred transaction types influence purchase behavior. It helped identify which coupon codes and payment modes drive higher sales volumes, guiding future promotional and payment strategy decisions.

Revenue by Transaction Bank (Bar Chart):
This visualization analyzed the total sales contributed by each transaction bank, offering insights into which banking partners facilitated the highest transaction volumes. It helped identify the most frequently used banks by customers, supporting decisions related to preferred payment partnerships and improving financial transaction efficiency.

Monthly and Daily Sales Trend (Line Chart):
This visualization tracked daily and monthly sales trends over the two-month period, highlighting fluctuations in purchasing activity. It helped identify peak sales days, seasonal patterns, and customer buying behavior over time, providing valuable insights for planning promotions and managing inventory effectively.

Sales Distribution by Payment Method (Donut Chart):

This visualization showed how sales were distributed across different payment methods, helping to understand customer preferences in transaction choices such as UPI, Credit Card, and Net Banking. During data preparation, null values in the Payment Mode column were replaced with "Unknown," but those records were excluded from this visual to focus on meaningful insights. The chart revealed the most commonly used payment methods, providing a clear understanding of how customers prefer to complete their purchases and supporting decisions to enhance payment-related strategies.

5. INSIGHTS & ANALYSIS

The sales dashboard provided a comprehensive view of company performance across two months, enabling deeper understanding of revenue distribution, customer activity, and sales channels.

The **Total Sales KPI** revealed the overall revenue generated during the period, serving as a benchmark for evaluating performance across sources and products. The **Unique Users KPI** highlighted the total number of individual customers, offering insights into user reach and engagement. Additionally, the **Average Sales per User** metric indicated variations in customer spending behavior across sales sources, identifying which channels attracted higher-value customers.

The **Sales Performance by Source** visualization showed that Direct sales contributed the highest share of total revenue, while Referral and Affiliate sources performed relatively lower. This suggests that focused marketing efforts on secondary channels could improve their contribution to overall sales.

The **Sales Distribution by Payment Method** revealed customer preferences toward specific payment types. After excluding the "Unknown" values, the analysis showed a strong inclination toward certain payment methods, helping the business understand which options to prioritize and promote.

The **Sales by Product Analysis** identified the top 10 revenue-generating products, offering a clear perspective on high-performing items that drive the majority of sales. This insight supports better inventory planning and targeted promotional strategies.

The **Transaction Bank Performance** visualization provided insight into how different banks contributed to total sales, indicating customer preferences toward specific banks or financial services.

The **Sales Trend by Day and Month** chart illustrated how daily sales fluctuated over the two-month period, revealing key patterns such as consistent performance peaks and potential low-activity periods. This aids in planning campaigns or resource allocation during slower days.

Finally, the **Repeat vs. One-Time Users** analysis highlighted the proportion of loyal customers making multiple purchases compared to new buyers, giving visibility into customer retention rates. Combined with the **Coupon and Payment Mode Matrix**, it provided valuable insights into how discounts and payment preferences influenced sales behavior.

6. RECOMMENDATIONS

Based on the insights derived from the dashboard, several strategic actions can be taken to enhance sales performance and customer engagement:

Enhance Customer Retention Programs:
The repeat user ratio indicates that only a small portion of customers return for additional purchases. Introducing loyalty programs, targeted email campaigns, and personalized offers can help improve retention.

Focus on High-Performing Products:
The top 10 products contribute significantly to total revenue. These products can be leveraged for promotional campaigns or bundled with lower-selling items to increase average order value.

Optimize Payment Options:
Analysis of payment modes (excluding "Unknown" data) shows customer preferences toward specific methods. Ensuring seamless transactions and adding popular payment gateways can further improve conversion rates.

Leverage Bank Partnerships:
Insights from transaction bank data suggest opportunities to collaborate with high-performing banks for cashback offers, EMI options, or exclusive discounts to attract more customers.

Monitor Seasonal and Daily Sales Trends:
The two-month sales trend indicates specific days with stronger performance. Scheduling promotions or ad campaigns around high-demand periods can maximize revenue potential.

Improve Data Collection Practices:
The presence of "Unknown" entries in fields like payment mode and coupon codes highlights gaps in data quality. Standardizing input methods or integrating automated data validation will enhance the reliability of future analyses.

7. CONCLUSION

This Power BI dashboard project successfully transformed raw transactional data into meaningful business insights through data cleaning, preparation, and visualization.
Key metrics such as **Total Sales**, **Unique Users**, and **Average Sales per User** provided a clear understanding of overall business performance, while advanced visuals like **Sales by Source**, **Product Performance**, and **Customer Retention** highlighted crucial sales and behavioral patterns.

By leveraging DAX measures, data transformations, and interactive visualizations, the analysis identified key revenue drivers, underperforming channels, and improvement areas. The findings support strategic decision-making in marketing, customer retention, and payment optimization.

Overall, the dashboard offers a comprehensive, data-driven view of sales performance across multiple dimensions, empowering the company to take informed, actionable steps toward growth and efficiency.

8. DASHBOARD SNAPSHOTS

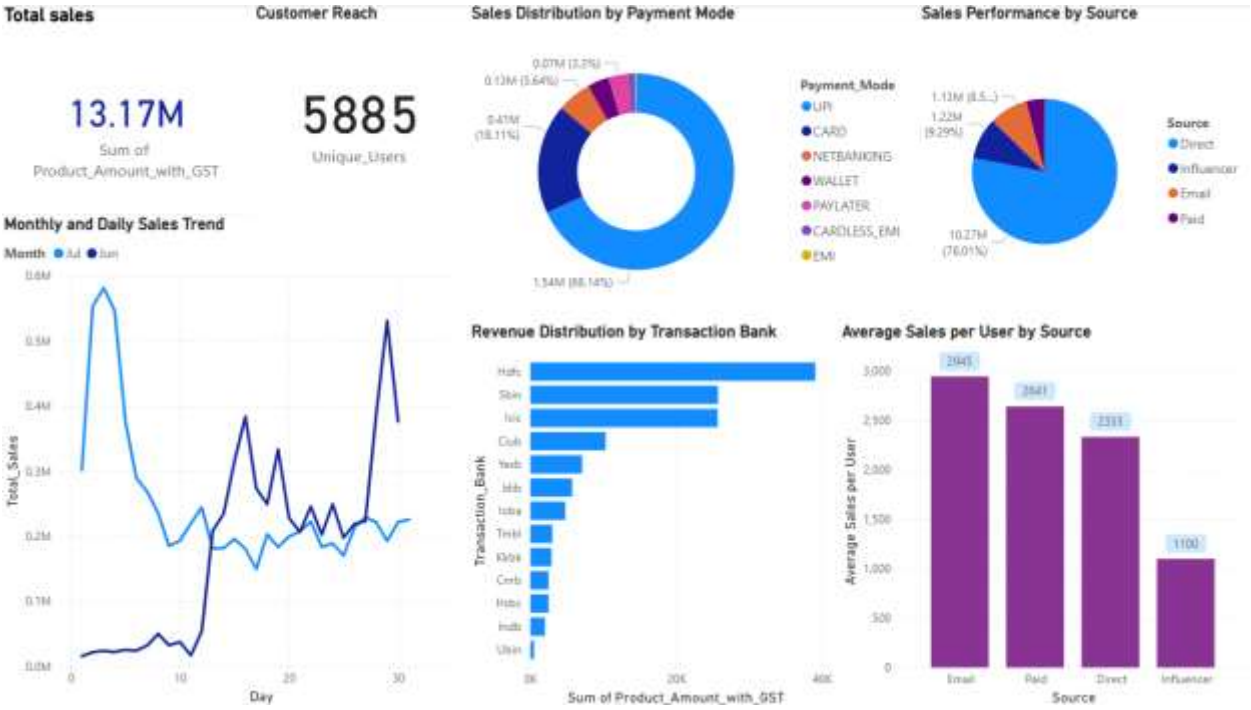


Figure 1: Sales and Customer Insights Dashboard

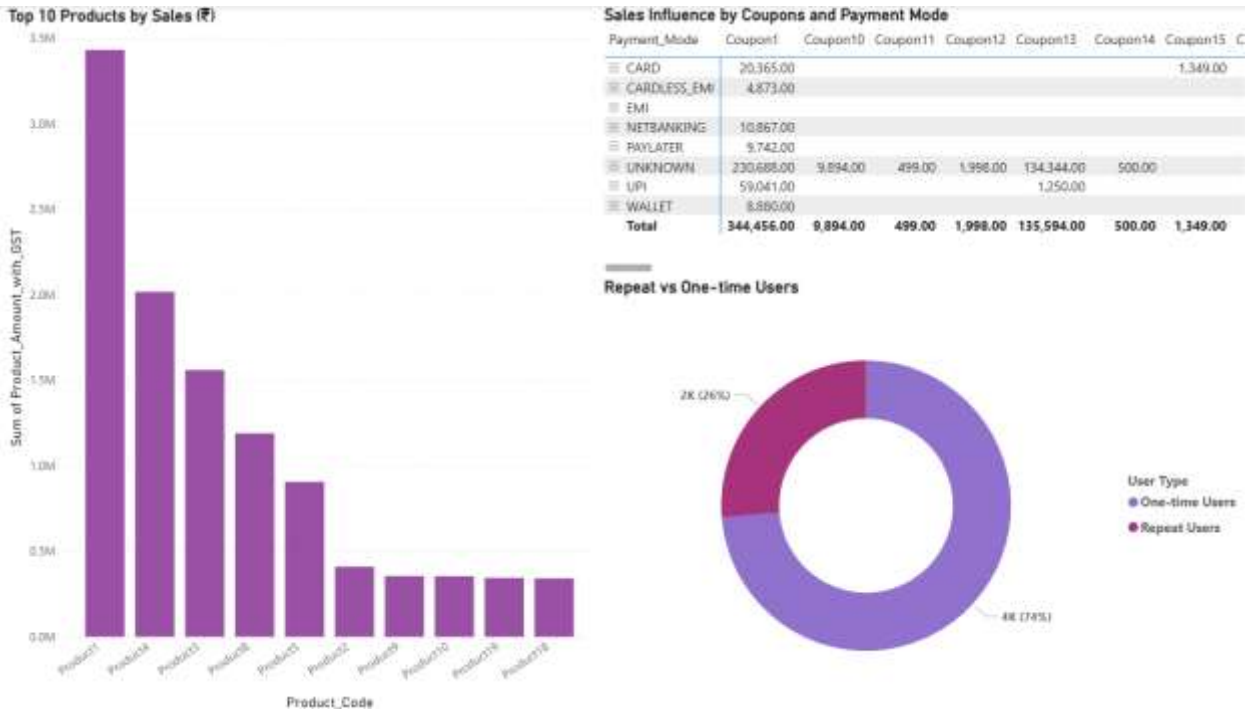


Figure 2: Product and User Behavior Analysis Dashboard